

The `coppe` document class

Version 3.8

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain
`\mainmatter` three-parts: *front*, *main* and *back* matters [?]. Each one of these parts is started
`\backmatter` by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`.
The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (`capa`)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The `capa` is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation [?]. It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your
`\foreigntitle` monograph in the current and foreign languages. The default languages are Brazilian Portuguese and English. The `babel` package is automatically loaded by `coppe.cls`, so you do not need to load it again. The Brazilian Portuguese is the main language and the English is only required for the foreign abstract.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc.
`\examiner`

students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the Academic Registry [?], abstracts must be in one page each, with
`foreignabstract (env.)` at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will be used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abbreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_EX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_EX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_EX styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
```

```
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, **msc**, **dscexam**, **mscexam** The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `mscexam`, and `dscexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

`english` COPPET_{EX} uses Babel. The default language is Portuguese (actually `brazilian`), with English being the second language. If option `english` is used, English becomes the main language and Portuguese the secondary. Look at the Babel package to switch between languages.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

6 Quick, useful tips

Pictures. The default picture format of L_AT_EX is the Encapsulated PostScript (EPS). If you use pdfL_AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_lat_ex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L_AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package. It is also useful to enable hyperlinks. Fortunately, the `coppe` class will do this automatically if `hyperref` is loaded.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

7 A simple example

```
1 <*example>
2 \documentclass[dsc]{coppe}
3
4 \usepackage{booktabs}% tabelas mais bonitas
5 \usepackage{rotating}% rodando coisas, como tabelas
6 \usepackage{longtable} % tabelas longas
7 \usepackage[most]{tcolorbox} % caixas de texto
8 \usepackage{tikz}          % gráficos vetoriais (o tcolorbox já o traz)
9 \usepackage{subcaption}    % subfiguras/subtabelas (NÃO use subfig/subfigure)
10 \usepackage{amsmath,amssymb}
11 \DeclareMathOperator{\RMSE}{RMSE}% operadores próprios (cap. de matemática)
12 \DeclareMathOperator{\diag}{diag}
13 \usepackage{hyperref}
14 % listings, fancyvrb e algorithm2e já vêm carregados pela classe coppe (v3.8).
15
16 \addbibresource{example.bib}% base de referências (biblatex/biber)
17
18 \makelossymbols
19 \makeloabbreviations
20
21 \begin{document}
22   \title{Título da Tese}
23   \foreigntitle{Thesis Title}
24   \author{Nome do Autor}{Sobrenome}
25   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
26   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
27   \advisor{Prof.}{Nome do Terceiro Orientador}{Sobrenome}{D.Sc.}
28
29   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
30   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
31   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
32   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
33   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
34   \department{PESC}
35   \date{01}{2024}
36
37   \keyword{Primeira palavra-chave}
38   \keyword{Segunda palavra-chave}
39   \keyword{Terceira palavra-chave}
40
41   \maketitle
42
43   \frontmatter
44   \dedication{A alguém cujo valor é digno desta dedicatória.}
45
46   \chapter*{Agradecimentos}
47
48   Gostaria de agradecer a todos os que contribu\iram para este trabalho:
49   orientadores, colegas, familiares e \as ag\ências de fomento. Lorem ipsum
50   dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
51   ut labore et dolore magna aliqua.
52
```

```

53 \begin{abstract}
54
55 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
56 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
57 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
58 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
59 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
60 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
61 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
62 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
63 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
64 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
65 aspernatur aut odit aut fugit.
66
67 \end{abstract}
68
69 \begin{foreignabstract}
70
71 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
72 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
73 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
74 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
75 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
76 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
77 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
78 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
79 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
80 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
81 aspernatur aut odit aut fugit.
82
83 \end{foreignabstract}
84
85 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
86 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
87 \listoffigures
88 \listoftables
89 \listofquadros
90 \listofprogramas
91 \listofalgorithms
92 \printloabbreviations
93 \printlosymbols
94 \tableofcontents
95
96 \mainmatter
97 \chapter{Introdução}
98
99 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
100
101 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
102
103 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
104
105 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
106

```


107 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
108
109 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX` para a A
110
111 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci
112
113 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
114
115 Este documento não substitui, mas complementa, o documento que descreve a classe.
116
117
118
119 `\chapter{Configurações Iniciais}`
120
121 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
122
123 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
124
125 Recomendo ler o documento “The `\verb*|coppe|` document class” para entender melhor todas
126
127 `\section{Linguagem principal do texto}`
128
129 Essa classe considera que o texto principal está em português e algumas partes específicas
130 `\begin{itemize}`
131 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.
132 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
133 `\end{itemize}`
134
135 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\Coppet`
136
137
138 `\section{Por que usar o \LaTeX}`
139
140 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess
141
142 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p
143
144 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
145
146 `\section{Como e onde usar o \LaTeX}`
147
148 Existem muitos tutoriais de `\LaTeX`, mas basicamente, em 2024, ele é usado em dois ambiente
149 `\begin{enumerate}`
150 `\item` Na sua máquina, instalando uma versão completa como o `\hologo{MiKTeX}`, típico do Win
151 `\item` Usar um ambiente na rede, como o `Overleaf`.
152 `\end{enumerate}`
153
154 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
155
156
157 `\chapter{Algumas Regras da COPPE}`
158
159 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
160

161 É imprescindível definir os símbolos, tal como o
162 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
163 `\syml{\mathbb{R}}{Conjunto dos números reais}`
164 `\syml{\emptyset}{Conjunto vazio}`. Usamos esse exemplo aqui justamente para mostrar como
165
166 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o `\ve`
167
168 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índices
169
170 `\section{Citações}`
171 `\label{sec:citacoes}`
172
173 Citações curtas podem ser feitas `\quote{o comando quote}` ou direto com “duas crases e dois
174
175 `\begin{longquote}`
176 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
177 feita com o ambiente `\verb=longquote=` The primary objective of this
178 investigation was to determine the feasibility of detecting corrosion in
179 aluminum Naval aircraft components with neutron radiographic interrogation
180 and the use of standard corrosion penetrameters. Secondary objectives
181 included the determination of the effect of object thickness on image quality,
182 the defining of minimum levels of detectability and a preliminary investigation
183 of a means whereby the degree of corrosion could be quantified with neutron
184 radiographic data. `\cite{article-example}`
185 `\end{longquote}`
186
187 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb*|na`
188
189 Em todo caso, `\textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em`
190
191 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de ref
192
193 `\chapter{Floats}`
194
195 Grande parte dos problemas de iniciantes, e veteranos, em $\text{\LaTeX}$ é da localização dos `\texti`
196
197 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
198
199 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|\caption|` `\textbf{de toda ilustração}`
200
201 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao
202
203
204 `\section{Tabelas e Figuras Padrão}`
205
206 Vamos ver uma tabela padrão, como a `\autoref{tab:exemplo_numeros}`.
207
208 `\begin{table}[ht]`
209 `\centering` % Centraliza a tabela
210 `\caption{Exemplo de Tabela de Números}`
211 `\label{tab:exemplo_numeros}`
212 `\begin{tabular}{ccc}` % Define a quantidade de colunas
213 `\hline` % Linha superior
214 `\textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\` % Cabeçalhos

```

215 \hline % Linha média
216 1 & 2 & 3 \\ % Primeira linha de dados
217 \hline
218 4 & 5 & 6 \\ % Segunda linha de dados
219 \hline
220 7 & 8 & 9 \\ % Terceira linha de dados
221 \hline
222 10 & 11 & 12 \\ % Quarta linha de dados
223 \hline % Linha inferior
224 \end{tabular}
225 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
226 \end{table}
227
228
229
230 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
231
232 \begin{figure}[ht]
233 \centering % Centraliza a figura
234 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
235 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
236 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
237 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
238 \end{figure}
239
240 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
241
242 \begin{quadro}[ht]
243 \centering
244 \caption{Exemplo de Quadro}
245 \label{qua:exemplo}
246 \begin{tabular}{|l|l|}
247 \hline
248 \textbf{Ilustração} & \textbf{Delimitação} \\
249 \hline
250 Tabela & Linhas apenas no topo e na base. \\
251 \hline
252 Quadro & Linhas em todas as bordas. \\
253 \hline
254 \end{tabular}
255 \fonte{Elabora\c{c}\~ao pr\'opria.}
256 \end{quadro}
257
258
259
260
261
262 \section{Tabelas mais elegantes}
263
264 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
265
266 \begin{table}[ht]
267 \centering % Centraliza a tabela
268 \caption{Exemplo de Tabela de Números mais elegantes}

```

```

269 \label{tab:exemplo_numerosbom}
270 \begin{tabular}{ccc} % Define a quantidade de colunas
271 \toprule % Linha superior
272 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
273 \midrule % Linha média
274 1 & 2 & 3 \\ % Primeira linha de dados
275 4 & 5 & 6 \\ % Segunda linha de dados
276 7 & 8 & 9 \\ % Terceira linha de dados
277 10 & 11 & 12 \\ % Quarta linha de dados
278 \bottomrule % Linha inferior
279 \end{tabular}
280 \fonte{Elabora\c{c}\~ao pr\'opria.}
281 \end{table}
282
283 \section{Tabelas Longas ou Largas}
284
285 Se sua tabela é muito longa ou larga, existem várias opções.
286 \begin{itemize}
287     \item alterar o tamanho da letra
288     \item Usar o longtable
289     \item rodar a tabela, fazendo ela em \textit{landscape}
290     \item fazer a tabela dentro de um minibox
291 \end{itemize}
292
293
294 \subsection{Tabelas largas demais}
295
296 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
297
298 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
299
300 \begin{table}[ht]
301 \centering % Centraliza a tabela
302 \caption{Exemplo de Tabela Larga com Fonte Menor}
303 \label{tab:tabela_largafns}
304 \footnotesize % Aplica uma fonte menor para a tabela
305 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
306 \toprule
307 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} & \textbf{Coluna 6} & \textbf{Coluna 7} \\
308 \midrule
309 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
310 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
311 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
312 \bottomrule
313 \end{tabular}
314 \fonte{Elabora\c{c}\~ao pr\'opria.}
315 \end{table}
316
317 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
318
319 \begin{table}[ht]
320 \centering
321 \caption{Exemplo de Tabela Redimensionada}
322 \label{tab:examplerb}

```

```

323 \resizebox{\textwidth}{!}{%
324 \begin{tabular}{llll}
325 \toprule
326 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
327 \midrule
328 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
329 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
330 \bottomrule
331 \end{tabular}%
332 }
333 \fonte{Elabora\c{c}\~ao pr\'opria.}
334 \end{table}
335
336
337 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\ea pode usar o pacote \verb*|rotat
338
339 Aqui est\aa um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
340
341 \begin{sidewaystable}
342 \centering
343 \caption{Sua Legenda Aqui}
344 \label{tab:sua_tabela}
345 \begin{tabular}{lll}
346 \toprule
347 Coluna 1 & Coluna 2 & Coluna 3 \\
348 \midrule
349 Item 1 & Item 2 & Item 3 \\
350 Item 4 & Item 5 & Item 6 \\
351 \bottomrule
352 \end{tabular}
353 \fonte{Elabora\c{c}\~ao pr\'opria.}
354 \end{sidewaystable}
355
356 Se a tabela for muito longa, o ambiente \verb|longtable| \e o ideal. Ele fornece comandos par
357
358 % Exemplo de tabela longa que se estende por v\arias p\aginas
359 \begin{longtable}{|c|c|c|}
360 % primeiro cabe\calo (e o caption)
361 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
362 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\
363 \endfirsthead
364 % cabe\calo normal
365 \multicolumn{3}{c}%
366 {\table\thetable} -- continua\c\o da p\agina anterior} \\
367 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\
368 \endhead
369 % p\e normal
370 \hline \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\
371 \endfoot
372 \hline
373 % \u\timo p\e
374 \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\
375 \hline \hline
376 \endlastfoot

```

```

377
378 % Conteúdo da tabela
379 1 & 2 & 3 \\
380 4 & 5 & 6 \\
381 1 & 2 & 3 \\
382 4 & 5 & 6 \\
383 1 & 2 & 3 \\
384 4 & 5 & 6 \\
385 1 & 2 & 3 \\
386 4 & 5 & 6 \\
387 1 & 2 & 3 \\
388 4 & 5 & 6 \\
389 1 & 2 & 3 \\
390 4 & 5 & 6 \\
391 1 & 2 & 3 \\
392 4 & 5 & 6 \\
393 1 & 2 & 3 \\
394 4 & 5 & 6 \\
395 1 & 2 & 3 \\
396 1 & 2 & 3 \\
397 4 & 5 & 6 \\
398 1 & 2 & 3 \\
399 4 & 5 & 6 \\
400 1 & 2 & 3 \\
401 4 & 5 & 6 \\
402 1 & 2 & 3 \\
403 4 & 5 & 6 \\
404 1 & 2 & 3 \\
405 4 & 5 & 6 \\
406 1 & 2 & 3 \\
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409 4 & 5 & 6 \\
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411 4 & 5 & 6 \\
412 1 & 2 & 3 \\
413 4 & 5 & 6 \\
414 1 & 2 & 3 \\
415 4 & 5 & 6 \\
416 1 & 2 & 3 \\
417 4 & 5 & 6 \\
418 1 & 2 & 3 \\
419 4 & 5 & 6 \\1 & 2 & 3 \\
420 4 & 5 & 6 \\
421 1 & 2 & 3 \\
422 4 & 5 & 6 \\
423 1 & 2 & 3 \\
424 1 & 2 & 3 \\
425 4 & 5 & 6 \\
426 1 & 2 & 3 \\
427 4 & 5 & 6 \\
428 1 & 2 & 3 \\
429 4 & 5 & 6 \\
430 1 & 2 & 3 \\

```

431 4 & 5 & 6 \\
 432 1 & 2 & 3 \\
 433 4 & 5 & 6 \\
 434 1 & 2 & 3 \\
 435 4 & 5 & 6 \\
 436 1 & 2 & 3 \\
 437 4 & 5 & 6 \\
 438 1 & 2 & 3 \\
 439 4 & 5 & 6 \\
 440 1 & 2 & 3 \\
 441 4 & 5 & 6 \\
 442 1 & 2 & 3 \\
 443 4 & 5 & 6 \\
 444 1 & 2 & 3 \\
 445 4 & 5 & 6 \\
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 448 1 & 2 & 3 \\
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 471 4 & 5 & 6 \\
 472 1 & 2 & 3 \\
 473 4 & 5 & 6 \\
 474 1 & 2 & 3 \\
 475 4 & 5 & 6 \\
 476 1 & 2 & 3 \\
 477 4 & 5 & 6 \\
 478 1 & 2 & 3 \\
 479 4 & 5 & 6 \\
 480 1 & 2 & 3 \\
 481 4 & 5 & 6 \\
 482 1 & 2 & 3 \\
 483 4 & 5 & 6 \\
 484 1 & 2 & 3 \\

```

485 4 & 5 & 6 \\
486
487 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
488 \end{longtable}
489 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
490
491 \section{Subfiguras (subcaption)}
492
493 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
494
495 \begin{figure}[ht]
496   \centering
497   \begin{subfigure}[b]{0.42\textwidth}
498     \centering\includegraphics[width=\linewidth]{example-image-a}
499     \caption{Primeira vista}\label{fig:sub-a}
500   \end{subfigure}
501   \hfill
502   \begin{subfigure}[b]{0.42\textwidth}
503     \centering\includegraphics[width=\linewidth]{example-image-b}
504     \caption{Segunda vista}\label{fig:sub-b}
505   \end{subfigure}
506   \caption{Figura com duas subfiguras}
507   \label{fig:subs}
508   \fonte{Elabora\c{c}\~ao pr\'opria.}
509 \end{figure}
510
511 Referencie a figura inteira com \verb|\autoref{fig:subs}| e cada parte com \verb|\autoref{fig:sub-a}|
512
513 \section{Imagens (graphicx)}
514
515 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
516 \begin{itemize}
517   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
518   \item \verb|[angle=90]| --- rotação;
519   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direita, topo});
520 \end{itemize}
521 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{eps}
522
523 \section{Introdução ao tikz}
524
525 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
526
527 \begin{figure}[ht]
528   \centering
529   \begin{tikzpicture}
530     \node[draw, rounded corners] (a) at (0,0) {Início};
531     \node[draw, rounded corners] (b) at (4,0) {Fim};
532     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
533   \end{tikzpicture}
534   \caption{Diagrama simples em tikz}
535   \label{fig:tikz}
536   \fonte{Elabora\c{c}\~ao pr\'opria.}
537 \end{figure}
538

```


539 Para gráficos de dados (curvas, barras), veja o `\verb|pgfplots|`, construído sobre o `\verb|tikz|`

540

541 `\section{Float ou direto no texto?}`

542

543 Colocar a tabela ou imagem dentro de `\verb|figure|`/`\verb|table|` (um `\emph{float}`) deixa o `\LaTeX`

544

545 `\chapter{Using BibLaTeX}`

546 `\label{cap:biblatex}`

547

548 Esta classe gera as referências com BibLaTeX e o `\emph{backend}` `\texttt{biber}`, no padrão

549

550 `\textbf{É obrigatório compilar com \texttt{biber}}` (e não com `\hologo{BibTeX}`). A sequência

551

552 Os comandos do `\texttt{natbib}` (`\verb|\cite|`, `\verb|\citep|`) continuam disponíveis porque

553

554 `\begin{table}[h]`

555 `\caption{Exemplos de citações utilizando o comando padrão`

556 `\texttt{\textbackslash cite} do \LaTeX e`

557 `o comando \texttt{\textbackslash cite},`

558 `disponível pela opção \texttt{natbib} do BibLaTeX.}`

559 `\label{tab:citation}`

560 `\centering`

561 `{\footnotesize`

562 `\begin{tabular}{|c|c|c|}`

563 `\hline`

564 `Tipo da Publicação & \verb|\cite| & \verb|\citep|\\`

565 `\hline`

566 `Livro & \cite{book-example} & \citep{book-example}\\`

567 `Artigo & \cite{article-example} & \citep{article-example}\\`

568 `Relatório & \cite{techreport-example} & \citep{techreport-example}\\`

569 `Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\`

570 `Anais de Congresso & \cite{inproceedings-example} &`

571 `\citep{inproceedings-example}\\`

572 `Séries & \cite{incollection-example} & \citep{incollection-example}\\`

573 `Em Livro & \cite{inbook-example} & \citep{inbook-example}\\`

574 `Dissertação de mestrado & \cite{mastersthesis-example} &`

575 `\citep{mastersthesis-example}\\`

576 `Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\`

577 `\hline`

578 `\end{tabular}}`

579 `\fonte{Elabora-se aqui o conteúdo da tabela.}`

580 `\end{table}`

581

582 `\begin{table}[h]`

583 `\caption{Exemplos de citações utilizando o comando padrão`

584 `\texttt{\textbackslash cite} do \LaTeX e`

585 `o comando \texttt{\textbackslash cite},`

586 `disponível pela opção \texttt{natbib} do BibLaTeX. Além disso, usando o booktabs.}`

587 `\label{tab:citation1}`

588 `\centering`

589 `{\footnotesize`

590 `\begin{tabular}{ccc}`

591 `\toprule`

592 `Tipo da Publicação & \verb|\cite| & \verb|\citep|\\`

```

593 \midrule
594 Livro & \cite{book-example} & \citet{book-example}\\
595 Artigo & \cite{article-example} & \citet{article-example}\\
596 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
597 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
598 Anais de Congresso & \cite{inproceedings-example} &
599 \citet{inproceedings-example}\\
600 Séries & \cite{incollection-example} & \citet{incollection-example}\\
601 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
602 Dissertação de mestrado & \cite{mastersthesis-example} &
603 \citet{mastersthesis-example}\\
604 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
605 \bottomrule
606 \end{tabular}}
607 \fonte{Elabora\c{c}\~ao pr\'opria.}
608 \end{table}
609
610 \section{Comandos de citação}
611
612 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
613
614 \subsection{Parâmetro opcional: página e “tradução nossa”}
615
616 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
617
618 \subsection{Citação de citação (\emph{apud})}
619
620 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citeapud| (demonstra
621
622 \subsection{Múltiplas bibliografias}
623
624 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
625
626 \section{Tipos de entrada bibliográfica}
627
628 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
629
630 \subsection{Tipos acadêmicos principais}
631 \begin{itemize}
632 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
633 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
634 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
635 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
636 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
637 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
638 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
639 \end{itemize}
640
641 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
642 \begin{itemize}
643 \item Patente --- \verb|@patent| (\verb|@patente|)
644 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|
645 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@partitura|)
646 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)

```

```

647 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|
648 \end{itemize}
649
650 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
651 \begin{itemize}
652 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
653 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadstv|, \verb|@serietv|)
654 \end{itemize}
655
656 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
657
658 \chapter{Alguns outros exemplo úteis}
659
660 \begin{tcolorbox}[title=Meu Textbox]
661 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
662 \end{tcolorbox}
663
664 \begin{tcolorbox}
665 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
666 \end{tcolorbox}
667
668 \begin{figure}[ht]
669 \centering
670 \begin{tikzpicture}
671 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{
672 \begin{scope}[x={(image.south east)},y={(image.north west)}]
673 % Definindo o textbox dentro da figura
674 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
675 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
676 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
677 \end{tcolorbox}
678 };
679 \end{scope}
680 \end{tikzpicture}
681 \caption{Figura com Textbox}
682 \label{fig:figura_com_textbox1}
683 \fonte{Elabora\c{c}\~ao pr\'opria.}
684 \end{figure}
685
686
687 \begin{figure}[ht]
688 \centering
689 \begin{tcolorbox}
690 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
691 \end{tcolorbox}
692 \caption{Figura com Textbox simples}
693 \label{fig:figura_com_textbox}
694 \fonte{Elabora\c{c}\~ao pr\'opria.}
695 \end{figure}
696
697 \section{Listagens de código e algoritmos}
698
699 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
700

```

```

701 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
702 def saudacao(nome):
703     # imprime uma saudação acentuada
704     print(f"Olá, {nome}!")
705
706 saudacao("COPPE")
707 \end{python}
708 \source{Elaboração própria.}
709
710 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
711
712 \begin{gxoutput}
713 Olá, COPPE!
714 \end{gxoutput}
715
716 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
717
718 \begin{algorithm}[H]
719 \caption{Soma dos elementos de um vetor}
720 \label{alg:soma}
721 \Dados{vetor  $v$  com  $n$  elementos}
722 \Resultado{soma  $s$  dos elementos de  $v$ }
723  $s \leftarrow 0$ ;
724 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
725      $s \leftarrow s + v[i]$ ;
726 }
727 \Retorna  $s$ ;
728 \end{algorithm}
729
730 \chapter{Escrevendo matemática}
731
732 A classe carrega \texttt{amsmath} e \texttt{amssymb}. Fórmulas no texto vão entre \verb|$.
733
734 \section{Alinhamento (align)}
735 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
736 \begin{align}
737     f(x) &= (x+1)^2 \\
738         &= x^2 + 2x + 1 .
739 \end{align}
740
741 \section{Matrizes (matrix)}
742 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
743 \begin{equation}
744     A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
745     \quad
746     \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
747 \end{equation}
748
749 \section{Subequações (subequations)}
750 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
751 \begin{subequations}
752 \begin{align}
753     a &= b + c , \\
754     d &= e - f .

```

```

755 \end{align}
756 \end{subequations}
757
758 \section{Estruturas alinhadas (array)}
759 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
760 \begin{equation}
761 |x| = \left\{ \begin{array}{l}
762 x, \quad \text{se } x \geq 0, \\
763 -x, \quad \text{caso contrário.}
764 \end{array} \right.
765 \end{equation}
766
767 \section{Operadores (\textbackslash DeclareMathOperator)}
768 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
769 \begin{verbatim}
770 \usepackage{amsmath}
771 \DeclareMathOperator{\RMSE}{RMSE}
772 \DeclareMathOperator{\diag}{diag}
773 \end{verbatim}
774 Assim, \verb|$ \RMSE(y, \hat{y})| produz  $\RMSE(y, \hat{y})$ , e \verb|$ \diag(\lambda_1, \dots
775
776 \section{Unicode e LuaLaTeX (unicode-math)}
777 Ao migrar para o \hologo{LuaLaTeX}, troque o \texttt{amssymb} pelo \texttt{unicode-math} e
778 \begin{verbatim}
779 \usepackage{amsmath}
780 \usepackage{unicode-math}
781 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
782 \end{verbatim}
783 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x) dx$ ,  $\alpha$ ) e você ai
784
785 \chapter{Referências cruzadas, notas e índices}
786 \label{cap:refs}
787
788 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
789
790 \section{Rótulos e referências cruzadas}
791
792 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
793
794 \section{Notas de rodapé}
795
796 Use \verb|\footnote{texto}| \footnote{Como esta nota, que a classe imprime em espaçamento s
797
798 \section{hyperref e marcadores}
799
800 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
801
802 \section{URLs}
803
804 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
805
806 \section{Índice remissivo}
807
808 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre

```

```

809
810 \section{Traduções (NBR 10520)}
811
812 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
813
814 \chapter{Truques de \LaTeX}
815
816 \section{Comentários mágicos (\texttt{\% !TeX})}
817 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
818 \begin{verbatim}
819 % !TeX program = lualatex
820 % !TeX encoding = UTF-8
821 % !TeX spellcheck = pt_BR
822 % !BIB program = biber
823 \end{verbatim}
824 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
825
826 \section{Espaço engolido}
827 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
828
829 \section{Escopo (grupos)}
830 Chaves \verb|\{...\}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
831
832 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash n
833 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
834
835 \section{\texttt{latexmkrc}, perl e Overleaf}
836 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
837
838 \chapter{Método Proposto}
839 \chapter{Resultados e Discussões}
840
841 \section{Algumas Demonstrações}
842
843 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
844 \syml[beta]{Beta}{A palavra Beta more e corrigida}
845 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
846 \syml[beta]{A palavra beta}
847 \syml[alpha]{A palavra alpha}
848 \syml[alpha]{Alpha}{A palavra Alpha}
849 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
850 \syml[marco]{Marco}{A palavra Marco corrigida}
851
852 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl
853 \abbrev{GoT}{Game of Thrones}
854 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
855 \abbrev[iot]{IoT}{IoT ordenado como iot}
856 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
857 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
858 \abbrev{IoT}{IoT com ordenação default}
859 \abbrev[ITU]{ITU}{ITU mesmo}
860
861 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
862 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}

```

```

863
864 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
865
866
867
868
869 \chapter{Conclusões}
870
871 \backmatter
872
873 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
874 % partitura --- declarados via sinônimos em português em example.bib.
875 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
876 \nocite{jurisprudencia-example}
877 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
878 \nocite{jornalcaderno-example,jornal-example}
879 \printbibliography
880
881
882
883 \appendix
884
885 \chapter{Um apêndice}
886
887 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
888
889 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
890
891
892 \chapter{Outro apêndice}
893
894 \annex
895
896
897 \chapter{Um Anexo}
898 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
899
900
901
902 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
903
904 No modelo \Coppetex os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
905
906
907 \chapter{Outro Anexo}
908
909
910 \end{document}
911
912 \</example>

```

8 Implementation

8.1 The ‘coppe.cls’ file

```
913 \*class
914 \def\filename{coppe.dtx}
915 \def\fileversion{v3.8}
916 \def\filedate{2026/05/25}
917 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
918 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
919 \LoadClass[12pt,a4paper,twoside]{book}
920 % The bibliography engine (biblatex+biber) is loaded further below, after babel,
921 % so the numbers class option can select the citation style at load time.
922 \RequirePackage{hyphenat}
923 \RequirePackage{lastpage}
924 \RequirePackage{ifthen}
925 \RequirePackage{graphicx}
926 \RequirePackage{setspace}
927 \RequirePackage{tabularx}
928 \RequirePackage{etoolbox}
929 \RequirePackage{eqparbox}
930 \RequirePackage{ltxcmds}
931 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
932 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
933 \def\TeX{\hologo{TeX}}
934 \def\LaTeX{\hologo{LaTeX}}
935 \def\LaTeXe{\hologo{LaTeXe}}
936 \def\BibTeX{\hologo{BibTeX}}
937 \def\pdfLaTeX{\hologo{pdfLaTeX}}
938 \def\pdfTeX{\hologo{pdfTeX}}
939 \def\LuaLaTeX{\hologo{LuaLaTeX}}
940 \def\XeLaTeX{\hologo{XeLaTeX}}
941 \def\MiKTeX{\hologo{MiKTeX}}
942 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
943 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
944 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
945 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
946 \RequirePackage{iftex}
947 \ifPDFTeX
948   \RequirePackage[utf8]{inputenc}
949   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
950     Although slower, you should try compiling with LuaLaTeX}}
951 \fi
952 \RequirePackage[T1]{fontenc}
953 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
954 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
955 \RequirePackage[a4paper,twoside,bindingoffset=1cm,%
956 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
957 \RequirePackage{titlesec}
958 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
959 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capítulo N").
960 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
961 % keep "N Title". \appendix and \annex raise the flag (see below).
962 \newif\if@coppeappendix \@coppeappendixfalse
```



```

963 \titleformat{\chapter}[hang]
964   {\normalfont\Large\bfseries}
965   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
966    \lex}{\MakeUppercase}
967 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
968 \let\coppe@orig@appendix\appendix
969 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
970 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
971 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
972 % which macro is written (so it is correct at .toc-render time even though the
973 % appendixname has changed back); the chapter letter is baked in at write time.
974 \newif\if@coppeannex \@coppeannexfalse
975 \newcommand\coppe@tocapp[1]{%
976   \MakeUppercase{\iflanguage{brazilian}{Ap\^endice}{Appendix}}}%
977   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
978 \newcommand\coppe@tocanx[1]{%
979   \MakeUppercase{\iflanguage{brazilian}{Anexo}{Annex}}}%
980   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
981 \def\@chapter[#1]#2{%
982   \ifnum \c@secnumdepth >\m@ne
983     \if@mainmatter
984       \refstepcounter{chapter}%
985       \typeout{\@chapapp\space\thechapter.}%
986       \if@coppeappendix
987         \if@coppeannex
988           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
989         \else
990           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
991         \fi
992       \else
993         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
994       \fi
995     \else
996       \addcontentsline{toc}{chapter}{#1}%
997     \fi
998   \else
999     \addcontentsline{toc}{chapter}{#1}%
1000   \fi
1001   \chaptermark{#1}%
1002   \addtocontents{lof}{\protect\addvspace{10\p@}}%
1003   \addtocontents{lot}{\protect\addvspace{10\p@}}%
1004   \if@twocolumn
1005     \@topnewpage[\@makechapterhead{#2}]%
1006   \else
1007     \@makechapterhead{#2}%
1008     \@afterheading
1009   \fi}
1010 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1011 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1012 \titleformat{name=chapter,numberless}[hang]
1013   {\normalfont\Large\bfseries\filcenter}{\0pt}{\MakeUppercase}
1014 \titlespacing*{name=chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1015 % section: ALL CAPS, normal weight.
1016 \titleformat{\section}

```

```

1017 {\normalfont\large\bfseries}{\thesection}{lex}{\MakeUppercase}
1018 % subsection: bold (author writes Title Case).
1019 \titleformat{\subsection}
1020 {\normalfont\bfseries}{\thesubsection}{lex}{}
1021 % subsubsection: bold italic.
1022 \titleformat{\subsubsection}
1023 {\normalfont\bfseries\itshape}{\thesubsubsection}{lex}{}
1024 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1025 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1026 \RequirePackage{caption}
1027 \captionsetup{font=footnotesize,labelsep=endash}
1028 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1029 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1030 % at the bottom is mandatory). The float package's plaintop style forces the
1031 % caption above the body for figure, table and the new quadro float. -----
1032 \RequirePackage{float}
1033 \floatstyle{plaintop}
1034 \restylefloat{figure}
1035 \restylefloat{table}
1036 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1037 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1038 % unlike a tabela (rules on top and bottom only).
1039 \providecommand{\quadroname}{\iflanguage{brazilian}{Quadro}{Frame}}
1040 \providecommand{\listquadroname}{%
1041 \iflanguage{brazilian}{Lista de Quadros}{List of Frames}}
1042 \newfloat{quadro}{htbp}{loq}[chapter]
1043 \floatname{quadro}{\quadroname}
1044 \providecommand{\quadroautorefname}{\quadroname}
1045 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1046 % (float's own \l@quadro lays the number/title out incorrectly).
1047 \let\l@quadro\l@figure
1048 \newcommand\listofquadros{%
1049 \chapter*{\listquadroname}%
1050 \addcontentsline{toc}{chapter}{\listquadroname}%
1051 \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1052 \@starttoc{loq}}
1053 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1054 % float as a smaller, single-spaced, centred line; it does NOT advance any
1055 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1056 % the label is "Fonte" (pt) / "Source" (en).
1057 \newcommand{\cpsourcenname}{\iflanguage{brazilian}{Fonte}{Source}}
1058 \newcommand{\cpsource}[1]{%
1059 \par\addvspace{\smallskipamount}%
1060 {\footnotesize\singlespacing\centering\cpsourcenname: #1\par}}
1061 \let\cpfonte\cpsource
1062 \@ifundefined{source}{\let\source\cpsource}{}%
1063 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1064 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1065 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1066 \newcommand{\newcoppefloat}[3]{%
1067 {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1068 \floatname{#1}{#2}%
1069 \expandafter\newcommand\csname listof#1s\endcsname{%
1070 \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%

```

```

1071 \mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1072 \starttoc{lo#1}}
1073 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1074 % (filete) from the left margin; footnotesize is the book default. -----
1075 \let\coppe@orig@footnotetext\@footnotetext
1076 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1077 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1078 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1079 % (manual2024 2.6 / proposta P24). -----
1080 \RequirePackage{enumitem}
1081 \newlist{alineas}{enumerate}{2}
1082 \setlist[alineas,1]{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1083 \setlist[alineas,2]{label=--,leftmargin=1.8em,itemsep=0pt}
1084 % --- Page numbering: arabic in the top OUTER corner on textual pages
1085 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1086 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1087 % class option). -----
1088 \RequirePackage{fancyhdr}
1089 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1090 \renewcommand{\headrulewidth}{0pt}}
1091 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1092 % number may leak onto them (e.g. the verso right after the cover).
1093 \renewcommand{\cleardoublepage}{%
1094 \clearpage
1095 \if@twoside
1096 \ifodd\c@page\else
1097 \hbox{}\thispagestyle{empty}\newpage
1098 \if@twocolumn\hbox{}\newpage\fi
1099 \fi
1100 \fi}
1101 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1102 p\kern-.025em e}}\kern-.08em
1103 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1104 \newboolean{maledoc}
1105 \setboolean{maledoc}{false}
1106 %
1107 % Class options.
1108 % If you are writing a text in English, you must turn ‘English’ on.
1109 % Otherwise, Portuguese is considered the main language.
1110 \newif\if@english\@englishfalse
1111 \DeclareOption{english}{\@englishtrue}
1112 \DeclareOption{msc}{%
1113 \newcommand{\@degree}{M.Sc.}
1114 \newcommand{\@degreename}{Mestrado}
1115 \newcommand{\local@degname}{Mestre}
1116 \newcommand{\foreign@degname}{Master}
1117 \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1118 \newcommand\foreign@doctype{Dissertation}
1119 }
1120 \DeclareOption{dsceexam}{%
1121 \newcommand{\@degree}{D.Sc.}
1122 \newcommand{\@degreename}{Doutorado}
1123 \newcommand{\local@degname}{Doutor}
1124 \newcommand{\foreign@degname}{Doctor}

```

```

1125 \setboolean{maledoc}{true}
1126 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1127 \newcommand\foreign@doctype{Qualifying Exam}
1128 }
1129 \DeclareOption{mscexam}{%
1130 \newcommand{\@degree}{M.Sc.}
1131 \newcommand{\@degreename}{Mestrado}
1132 \newcommand{\local@degname}{Mestre}
1133 \newcommand{\foreign@degname}{Master}
1134 \setboolean{maledoc}{true}
1135 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1136 \newcommand\foreign@doctype{Qualifying Exam}
1137 }
1138 \DeclareOption{dsc}{%
1139 \newcommand{\@degree}{D.Sc.}
1140 \newcommand{\@degreename}{Doutorado}
1141 \newcommand{\local@degname}{Doutor}
1142 \newcommand{\foreign@degname}{Doctor}
1143 \newcommand\local@doctype{Tese}
1144 \newcommand\foreign@doctype{Thesis}
1145 }
1146 \newif\if@coppnumeric\@coppnumericfalse
1147 \DeclareOption{numbers}{\@coppnumerictrue}
1148 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1149 % pre-textual pages are counted but not numbered (manual2024 2.7).
1150 \newif\if@copperoman\@copperomanfalse
1151 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Here is the default one-and-a-half line spacing. Users can change to double spacing by passing the doublespacing option.

```

1152 \onehalfspacing
1153 \DeclareOption{doublespacing}{%
1154 \doublespacing
1155 }
1156 \ProcessOptions\relax
1157 \if@english
1158 \RequirePackage[brazilian,english]{babel}
1159 \else
1160 \RequirePackage[english,brazilian]{babel}
1161 \fi
1162 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1163 % Portuguese alias for short, inline quotations (proposta P3).
1164 \RequirePackage{csquotes}
1165 \newcommand{\citacao}[1]{\enquote{#1}}
1166 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1167 % package gives the "Programa" listing (caption on top, ABNT R46) with
1168 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1169 % the language styles below are based on the user's listagens.tex. -----
1170 \RequirePackage{xcolor}
1171 \RequirePackage{listings}
1172 \RequirePackage{fancyvrb}
1173 \definecolor{deepblue}{rgb}{0,0,0.5}
1174 \definecolor{deepred}{rgb}{0.6,0,0}
1175 \definecolor{deepgreen}{rgb}{0,0.5,0}

```

```

1176 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1177 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1178 \newcommand{\gxnumbersep}{0.5em}
1179 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1180   literate=
1181   {\~{\~{a}}1 {\~{\~{a}}1 {\~{\~{a}}1 {\~{\~{a}}1 {\~{\~{e}}1%
1182   {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1%
1183   {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1%
1184   {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1%
1185   {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1 {\~{\~{e}}1%
1186   {\~{\~{e}}1%
1187 }
1188 % Babel-aware listing names, set after listings' own babel setup.
1189 \AtBeginDocument{%
1190   \renewcommand{\lstlistingname}{\iflanguage{brazilian}{Programa}{Program}}%
1191   \renewcommand{\lstlistlistingname}{
1192     \iflanguage{brazilian}{Lista de Programas}{List of Programs}}%
1193 % \listofprogramas in the coppe list style (heading + ToC + running head).
1194 \newcommand{\listofprogramas}{
1195   \chapter*{\lstlistlistingname}%
1196   \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1197   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1198   \@starttoc{lol}}
1199 % Per-language styles + environments (python, java, xml, html, prolog).
1200 \newcommand{\pythonstyle}{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1201   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
1202   morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
1203   emph={MyClass,__init__},emphstyle=\ttb\color{deepred},%
1204   stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
1205   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1206   escapechar=',postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
1207 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
1208   \csname\@lst @SetFirstNumber\endcsname\csname\@lst @SaveFirstNumber\endcsname}
1209 \newcommand{\pythoninline}[1][\pythonstyle\lstinline!#1!}%
1210 \newcommand{\pythonexternal}[2][\pythonstyle\lstinputlisting[1]{#2}%
1211 \newcommand{\xmlstyle}{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1212   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
1213   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1214   keywordstyle=\ttb\color{deepblue},%
1215   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1216   emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
1217   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1218   postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
1219 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{
1220 \newcommand{\htmlstyle}{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1221   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
1222   keywordstyle=\ttb\color{deepblue},emphstyle=\ttb\color{deepred},%
1223   stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
1224   extendedchars=true,inputencoding=utf8,breaklines=true,%
1225   postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}
1226 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{
1227 \newcommand{\prologstyle}{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1228   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
1229   keywordstyle=\ttb\color{deepblue},emph={MyClass,__init__},%

```

```

1230 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
1231 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1232 breaklines=true,escapechar=',%
1233 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1234 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
1235 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1236 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1237 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
1238 morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
1239 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
1240 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1241 breaklines=true,escapechar=',%
1242 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1243 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
1244 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1245 \newcommand\javainline[1][\javastyle\lstinline!#1!}%
1246 \newcommand\javaexternal[2][\javastyle\lstinputlisting{#1}{#2}}
1247 % Verbatim output environments (program output), fancyvrb.
1248 \DefineVerbatimEnvironment{gxoutput}{Verbatim}%
1249 {frame=lines,numbers=left,numbersep=\gxnumbersep}
1250 \DefineVerbatimEnvironment{gxoutputs}{Verbatim}%
1251 {frame=lines,numbers=left,numbersep=\gxnumbersep,fontsize=\small}
1252 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1253 % Algorithms: algorithm2e, language following the english class option.
1254 \if@english
1255 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1256 \else
1257 \RequirePackage[portuguese,linesnumbered,lined,algochapter,ruled]{algorithm2e}
1258 \fi
1259 \SetKwFor{Para}{para}{faça}{fim do para}
1260 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1261 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
1262 % former natbib+bibtex setup. Loaded here, after babel, so its language strings
1263 % map correctly. The numbers option selects the numeric style; otherwise the
1264 % author-date default is used. natbib=true keeps \citel/\citep available.
1265 \if@coppenumeric
1266 \RequirePackage[backend=biber,database=coppe,%
1267 bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
1268 \else
1269 \RequirePackage[backend=biber,database=coppe,%
1270 bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
1271 \fi
1272 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
1273 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
1274 % "Bibliografia" (babel's default \bibname). The title is set directly with
1275 % \iflanguage so it is correct in BOTH the chapter heading and the .toc entry:
1276 % \iflanguage resolves to plain text and survives the write to the .toc, whereas
1277 % biblatex's \bibstring does not, and babel's \bibname caption override proved
1278 % unreliable here. Evaluated in the body, the current language is settled.
1279 \defbibheading{bibliography}{%
1280 \chapter*{\iflanguage{brazilian}{Refer\~encias}{References}}%
1281 \addcontentsline{toc}{chapter}{\iflanguage{brazilian}{Refer\~encias}{References}}}
1282 % The catalog page range (bib:begin/bib:end) was written by the old
1283 % thebibliography redefinition, which biblatex's \printbibliography does not

```

```

1284 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
1285 % begin label uses that hook and the end label is emitted by wrapping
1286 % \printbibliography to write it just after the reference list.
1287 \AtBeginBibliography{\coppe@bibBegin}
1288 \let\coppe@orig@printbibliography\printbibliography
1289 \renewcommand\printbibliography[1][]{\%
1290   \coppe@orig@printbibliography[#1]\%
1291   \coppe@bibEnd}

```

\department This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1292 \newcommand\department[1]{\%
1293   \ifthenelse{\equal{#1}{PEB}}{
1294     {\global\def\local@deptname{Engenharia Biom(\` e)dica}
1295     \global\def\foreign@deptname{Biomedical Engineering}}{}
1296   \ifthenelse{\equal{#1}{PEC}}{
1297     {\global\def\local@deptname{Engenharia Civil}
1298     \global\def\foreign@deptname{Civil Engineering}}{}
1299   \ifthenelse{\equal{#1}{PEE}}{
1300     {\global\def\local@deptname{Engenharia El(\` e)trica}
1301     \global\def\foreign@deptname{Electrical Engineering}}{}
1302   \ifthenelse{\equal{#1}{PEM}}{
1303     {\global\def\local@deptname{Engenharia Mec(\` a)nica}
1304     \global\def\foreign@deptname{Mechanical Engineering}}{}
1305   \ifthenelse{\equal{#1}{PEMM}}{
1306     {\global\def\local@deptname{Engenharia Metal(\` u)rgica e de Materiais}
1307     \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1308   \ifthenelse{\equal{#1}{PEN}}{
1309     {\global\def\local@deptname{Engenharia Nuclear}
1310     \global\def\foreign@deptname{Nuclear Engineering}}{}
1311   \ifthenelse{\equal{#1}{PEN0}}{
1312     {\global\def\local@deptname{Engenharia Oce(\` a)nica}
1313     \global\def\foreign@deptname{Ocean Engineering}}{}
1314   \ifthenelse{\equal{#1}{PPE}}{
1315     {\global\def\local@deptname{Planejamento Energ(\` e)tico}
1316     \global\def\foreign@deptname{Energy Planning}}{}
1317   \ifthenelse{\equal{#1}{PEP}}{
1318     {\global\def\local@deptname{Engenharia de Produ(c)(\` a)o}
1319     \global\def\foreign@deptname{Production Engineering}}{}
1320   \ifthenelse{\equal{#1}{PEQ}}{
1321     {\global\def\local@deptname{Engenharia Qu(\` i)mica}
1322     \global\def\foreign@deptname{Chemical Engineering}}{}
1323   \ifthenelse{\equal{#1}{PESC}}{
1324     {\global\def\local@deptname{Engenharia de Sistemas e Computa(c)(\` a)o}
1325     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1326   \ifthenelse{\equal{#1}{PET}}{
1327     {\global\def\local@deptname{Engenharia de Transportes}
1328     \global\def\foreign@deptname{Transportation Engineering}}{}
1329   \ifthenelse{\equal{#1}{PENT}}{
1330     {\global\def\local@deptname{Engenharia de Nanotecnologia}
1331     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1332 }

```

`\title` Used to enter the title in Brazilian Portuguese.

```
1333 \renewcommand\title[1]{%
1334   \global\def\local@title{#1}%
1335 }
```

`\foreigntitle` Used to enter the foreign title.

```
1336 \newcommand\foreigntitle[1]{%
1337   \global\def\foreign@title{#1}%
1338 }
```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```
1339 \newcount\@advisor\@advisor0
1340 \newcommand\advisor[4]{%
1341   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1342   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1343   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1344   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1345   \global\advance\@advisor by 1
1346   \ifnum\@advisor>1
1347     \renewcommand\local@advisorstring{Orientadores}
1348     \renewcommand\foreign@advisorstring{Advisors}
1349   \fi
1350 }
```

`\examiner`

```
1351 \newcount\@examiner\@examiner0
1352 \newcommand\examiner[3]{%
1353   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1354   \global\advance\@examiner by 1
1355 }
```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
1356 \renewcommand\author[2]{%
1357   \global\def\@authname{#1}
1358   \global\def\@authsurn{#2}
1359 }
```

`\date` This code makes easy to switch from dates in different languages.

```
1360 \renewcommand\date[2]{%
1361   \month=#1
1362   \year=#2
1363 }
```

`\local@monthname`

```
1364 \newcommand\local@monthname{\ifcase\month\or
1365   Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
1366   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
1367 \newcommand\foreign@monthname{\ifcase\month\or
1368   January\or February\or March\or April\or May\or June\or
1369   July\or August\or September\or October\or November\or December\fi}
```


\keyword

```
1370 \newcounter{keywords}
1371 \newcommand\keyword[1]{%
1372   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1373   \global\addtocounter{keywords}{1}
1374 }
```

\freeconfig This command allows easy changing of core class parameters.

```
1375 \newcommand\freeconfig[6]{%
1376   \providecommand\@degree{}
1377   \renewcommand\@degree{#1}
1378   \providecommand\@degreename{}
1379   \renewcommand\@degreename{#2}
1380   \providecommand\local@degname{}
1381   \renewcommand\local@degname{#3}
1382   \providecommand\foreign@degname{}
1383   \renewcommand\foreign@degname{#4}
1384   \providecommand\local@doctype{}
1385   \renewcommand\local@doctype{#5}
1386   \providecommand\foreign@doctype{}
1387   \renewcommand\foreign@doctype{#6}%
1388 }
1389 % \end{macrocode}
1390 % \end{macro}
1391 %
1392 % \begin{macro}{\frontmatter}
1393 % The number of pages for both frontmatter and mainmatter printed
1394 % in the cataloging details page is computed by means of simple
1395 % \LaTeX\ labels.
1396 % \begin{macrocode}
1397 \renewcommand\frontmatter{%
1398   \cleardoublepage
1399   \@mainmatterfalse
1400   \pagenumbering{roman}
1401   \thispagestyle{empty}
1402   % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1403   % start keeps the page-counter parity aligned with the physical (recto) side,
1404   % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1405   \setcounter{page}{1}
1406   \makefrontpage
1407   \clearpage
1408   \if@copperoman
1409     \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1410     \renewcommand{\headrulewidth}{0pt}}%
1411   \pagestyle{coppe}%
1412   \else
1413     \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1414     \pagestyle{empty}%
1415   \fi
1416 }
```

\mainmatter

```
1417 \renewcommand\mainmatter{%
1418   \coppe@mainBegin
```

```

1419 \cleardoublepage
1420 \@mainmattertrue
1421 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1422 \renewcommand{\headrulewidth}{0pt}}%
1423 \pagestyle{coppe}%
1424 \pagenumbering{arabic}}

\backmatter
1425 \renewcommand\backmatter{%
1426 \if@openright
1427 \cleardoublepage
1428 \else
1429 \clearpage
1430 \fi}
1431 %

\makecover The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is
the institution block — the university, the full COPPE name, and the graduate
program taken from \department — followed by the author, the title, and the
city and year, all centred and uppercased. The optional UFRJ and COPPE logos
head the page. The capa is not counted: it precedes the (roman) front matter and
is produced before the titlepage (folha de rosto).

1432 \newcommand\makecover{%
1433 \begin{titlepage}%
1434 \centering
1435 \makebox[\linewidth]{%
1436 \includegraphics[height=2cm]{ufrj-logo}\hfill
1437 \includegraphics[height=2cm]{coppe-logo}}\par
1438 \vspace{1.5cm}
1439 {\singlespacing
1440 \MakeUppercase{\local@universityname}\par
1441 \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de
1442 P{\`o}s-Gradua{\c c}{\`a}o e Pesquisa de Engenharia (COPPE)}}\par
1443 \MakeUppercase{Programa de \local@deptname}\par}
1444 \vspace{\stretch{2}}
1445 \textbf{\MakeUppercase{\@authname\ \@authsur}}\par
1446 \vspace{\stretch{2}}
1447 \nohyphens{%
1448 \if@english
1449 \MakeUppercase\foreign@title
1450 \else
1451 \MakeUppercase\local@title
1452 \fi}\par
1453 \vspace{\stretch{3}}
1454 \MakeUppercase{\local@cityname}\par
1455 \number\year
1456 \end{titlepage}%
1457 }

\maketitle
1458 \renewcommand\maketitle{%
1459 \pagenumbering{alph}
1460 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}}%

```

```

1461 \makecover
1462 \begin{titlepage}
1463 \begin{flushleft}
1464 \vspace*{1.5mm}
1465 \setlength\baselineskip{0pt}
1466 \setlength\parskip{1mm}
1467 \makebox[\linewidth]{%
1468   \includegraphics[height=2cm]{ufrj-logo}\hfill
1469   \includegraphics[height=2cm]{coppe-logo}}
1470 \end{flushleft}
1471 \vspace{1.05cm}
1472 \begin{center}
1473 \nohyphens{%
1474   \if@english
1475     \MakeUppercase\foreign@title
1476   \else
1477     \MakeUppercase\local@title
1478   \fi}\par
1479 \vspace*{3cm}
1480 \nohyphens{\@authname\ \@authsurn}\par
1481 \end{center}
1482 \vspace*{2.1cm}
1483 \begin{flushright}
1484 \begin{minipage}{8.45cm}
1485 \frontcover@maintext
1486 \end{minipage}\par
1487 \vspace*{7.5mm}
1488 \nohyphens{%
1489 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1490   \local@advisorstring: &
1491   \count1=0
1492   \toks@={}
1493   \@whilenum \count1<\@advisor \do{%
1494     \ifcase\count1 % same as \ifnum0=\count1
1495       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1496         \expandafter\endcsname\expandafter\space%
1497         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1498     \else
1499       \toks@=\expandafter\expandafter\expandafter{%
1500         \expandafter\the\expandafter\toks@%
1501         \expandafter&\expandafter\space%
1502         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1503         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1504       }%
1505     \fi
1506     \advance\count1 by 1}
1507   \the\toks@
1508 \end{tabularx}}\par
1509 \end{flushright}
1510 \vspace*{\fill}
1511 \begin{center}
1512 \local@cityname\par
1513 \local@monthname\ de \number\year
1514 \end{center}

```

```

1515 \end{titlepage}
1516 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1517 % the title page is a recto, so this lands on the immediately following verso,
1518 % ahead of the approval sheet produced by \frontmatter.
1519 \ifthenelse{\boolean{maledoc}}{\thispagestyle{empty}\makecatalog}%
1520 \global\let\maketitle\relax%
1521 \global\let\and\relax}

1522 \newcommand\makefrontpage{%
1523 \begin{center}
1524 \sloppy\nohyphens{
1525 \if@english
1526 \MakeUppercase\foreign@title
1527 \else
1528 \MakeUppercase\local@title
1529 \fi}\par
1530 \vspace*{7mm}
1531 {\@authname\ \@authsur}\par
1532 \end{center}\par
1533 \vspace*{4mm}
1534 \frontpage@maintext
1535 \vspace*{16mm}
1536 \nohyphens{%
1537 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
1538 \local@advisorstring: &
1539 \count1=0
1540 \toks@={}
1541 \@whilenum \count1<\@advisor \do{%
1542 \ifcase\count1 % same as \ifnum0=\count1
1543 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1544 \expandafter\endcsname\expandafter\space%
1545 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1546 \else
1547 \toks@=\expandafter\expandafter\expandafter{%
1548 \expandafter\the\expandafter\toks@%
1549 \expandafter&\expandafter\space%
1550 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1551 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1552 }%
1553 \fi
1554 \advance\count1 by 1}
1555 \the\toks@
1556 \end{tabularx}\par
1557 \vspace*{20mm}
1558 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
1559 Aprovada por: &
1560 \count1=0
1561 \toks@={}
1562 \@whilenum \count1<\@examiner \do{%
1563 \ifcase\count1 % same as \ifnum0=\count1
1564 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1565 \expandafter\endcsname\expandafter\\}
1566 \else
1567 \toks@=\expandafter\expandafter\expandafter{%

```

```

1568         \expandafter\the\expandafter\toks@%
1569         \expandafter&\expandafter\space%
1570         \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1571         \expandafter\space\\
1572     }%
1573 \fi
1574 \advance\count1 by 1}
1575 \the\toks@
1576 \end{tabularx}}\par
1577 \vspace*{\fill}
1578 \frontpage@bottomtext}

1579 \newcommand\coppe@hypersetup{%
1580 \begingroup
1581 % changes to \toks@ and \count@ are kept local;
1582 % it's not necessary for them, but it is usually the case
1583 % for \count1, because the first ten counters are written
1584 % to the DVI file, thus you got lucky because of PDF output
1585 \toks@={}% in this special case not necessary
1586 \count@=0 %
1587 \@whilenum\count@<\value{keywords}\do{%
1588 % * a keyword separator is not necessary,
1589 %   if there is just one keyword
1590 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1591 %   at least once, to get rid of the loop depended \count@
1592 \ifcase\count@ % same as \ifnum0=\count@
1593     \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1594 \else
1595     \toks@=\expandafter\expandafter\expandafter{%
1596         \expandafter\the\expandafter\toks@
1597         \expandafter;\expandafter\space
1598         \csname CoppeKeyword:\the\count@\endcsname
1599     }%
1600 \fi
1601 \advance\count@ by 1 %
1602 }%
1603 \edef\x{\endgroup
1604     \noexpand\hypersetup{%
1605         pdfkeywords={\the\toks@}%
1606     }%
1607 }%
1608 \x
1609 \hypersetup{%
1610     pdfauthor={\@authname\ \@authsurn},
1611     pdftitle={\local@title},
1612     pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
1613     pdfcreator={LaTeX with CoppeTeX toolkit},
1614     breaklinks={true},
1615     raiselinks={true},
1616     pageanchor={true},
1617 }}

```

\makecatalog When the document has illustrations, it is required to insert “: il;” between the number of pages of the textual part and the page dimension. We have created a

label to flag the existence of lists of figures. It is checked to be undefined using the plain $\text{\TeX}$ command $\text{\@isundefined}$ [?].

```

1618 \newcommand\makecatalog{%
1619   \vspace*{\fill}
1620   \begin{center}
1621     \setlength{\fboxsep}{5mm}
1622     \framebox[120mm][c]{\makebox[5mm][c]{}%
1623       \begin{minipage}[c]{105mm}
1624         \setlength{\parindent}{5mm}
1625         \noindent\sloppy\nohyphens\@authsurn,
1626         \nohyphens\@authname\par
1627         \nohyphens{%
1628           \if@english
1629             \foreign@title%
1630           \else
1631             \local@title%
1632           \fi/\@authname\ \@authsurn. -- \local@cityname:
1633   UFRJ/COPPE, \number\year.}\par
1634   \pageref{front:pageno},
1635   \pageref{LastPage}
1636   p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.}\par
1637   % There is an issue here. When the last entry must be split between lines,
1638   % the spacing between it and the next paragraph becomes smaller.
1639   % Should we manually introduce a fixed space? But how could we know that
1640   % a name was split? Is this happening yet?
1641   \nohyphens{%
1642   \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1643     \local@advisorstring: &
1644     \count1=0
1645     \toks@={}
1646     \@whilenum \count1<\@advisor \do{%
1647       \ifcase\count1 % same as \ifnum0=\count1
1648         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1649           \expandafter\endcsname\expandafter\space%
1650           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1651       \else
1652         \toks@=\expandafter\expandafter\expandafter{%
1653           \expandafter\the\expandafter\toks@
1654           \expandafter\&\expandafter\space
1655           \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1656           \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1657       }%
1658     \fi
1659     \advance\count1 by 1}
1660     \the\toks@
1661   \end{tabularx}}\par
1662   \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1663   UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
1664   Refer{\^ e}ncias Bibliogr{\` a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1665   \count1=0
1666   \count2=1
1667   \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1668     \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1669     \advance\count1 by 1

```

```

1670         \advance\count2 by 1}
1671     I. \csname CoppeAdvisorSurname:0\endcsname,%
1672     \ \csname CoppeAdvisorName:0\endcsname%
1673     \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}).
1674     II. \local@universityname, COPPE, Programa de \local@deptname.
1675     III. T{\ ' i}tulo.}
1676 \end{minipage}}
1677 \end{center}
1678 \vspace*{\fill}}

\dedication

1679 \newcommand\dedication[1]{
1680 \gdef\@dedic{#1}
1681 \cleardoublepage
1682 \vspace*{\fill}
1683 \begin{flushright}
1684 \begin{minipage}{60mm}
1685 \raggedleft \it \normalsize \@dedic
1686 \end{minipage}
1687 \end{flushright}}

abstract (env.) This is a specialization of the abstract in the article standard class.

1688 \newenvironment{abstract}{%
1689 \cleardoublepage
1690 \thispagestyle{plain}
1691 \abstract@toptext\par
1692 \vspace*{8.6mm}
1693 \begin{center}
1694 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1695 \vspace*{13.2mm}
1696 \@authname\ \@authsurn \par
1697 \vspace*{7mm}
1698 \local@monthname/\number\year
1699 \end{center}\par
1700 \vspace*{1.5cm}
1701 \noindent%
1702 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1703 \local@advisorstring: &
1704 \count1=0
1705 \toks@={}
1706 \@whilenum \count1<\@advisor \do{%
1707 \ifcase\count1 % same as \ifnum0=\count1
1708 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1709 \expandafter\endcsname\expandafter\space%
1710 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1711 \else
1712 \toks@=\expandafter\expandafter\expandafter{%
1713 \expandafter\the\expandafter\toks@
1714 \expandafter&\expandafter\space
1715 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1716 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1717 }%
1718 \fi
1719 \advance\count1 by 1}

```

```

1720 \the\toks@
1721 \end{tabularx}\par
1722 \vspace*{2mm}
1723 \noindent\local@deptstring: \local@deptname\par
1724 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1725 \newenvironment{foreignabstract}{%
1726 \cleardoublepage
1727 \thispagestyle{plain}
1728 \begin{otherlanguage}{english}
1729 \foreignabstract@toptext\par
1730 \vspace*{8.6mm}
1731 \begin{center}
1732 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1733 \vspace*{13.2mm}
1734 \@authname\ \@authsur \par
1735 \vspace*{7mm}
1736 \foreign@monthname/\number\year
1737 \end{center}\par
1738 \vspace*{1.5cm}
1739 \noindent%
1740 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1741 \foreign@advisorstring: &
1742 \count1=0
1743 \toks@={}
1744 \@whilenum \count1<\@advisor \do{%
1745 \ifcase\count1 % same as \ifnum0=\count1
1746 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1747 \expandafter\endcsname\expandafter\space%
1748 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1749 \else
1750 \toks@=\expandafter\expandafter\expandafter{%
1751 \expandafter\the\expandafter\toks@
1752 \expandafter&\expandafter\space
1753 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1754 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1755 }%
1756 \fi
1757 \advance\count1 by 1}
1758 \the\toks@
1759 \end{tabularx}\par
1760 \vspace*{2mm}
1761 \noindent\foreign@deptstring: \foreign@deptname\par
1762 \vspace*{7mm}}{%
1763 \end{otherlanguage}
1764 \vspace*{\fill}
1765 \global\let\@author\@empty
1766 \global\let\@date\@empty
1767 \global\let\foreign@title\@empty
1768 \global\let\foreign@title\relax
1769 \global\let\local@title\@empty
1770 \global\let\local@title\relax
1771 \global\let\author\relax

```



```

1772 \global\let\author\relax
1773 \global\let\date\relax}

\listoffigures
1774 \renewcommand\listoffigures{%
1775     \coppe@hasLof
1776     \if@twocolumn
1777         \@restonecoltrue\onecolumn
1778     \else
1779         \@restonecolfalse
1780     \fi
1781     \chapter*{\listfigurename}%
1782     \addcontentsline{toc}{chapter}{\listfigurename}%
1783     \@mkboth{\MakeUppercase\listfigurename}%
1784             {\MakeUppercase\listfigurename}%
1785     \@starttoc{lof}%
1786     \if@restonecol\twocolumn\fi
1787 }

\listoftables
1788 \renewcommand\listoftables{%
1789     \if@twocolumn
1790         \@restonecoltrue\onecolumn
1791     \else
1792         \@restonecolfalse
1793     \fi
1794     \chapter*{\listtablename}%
1795     \addcontentsline{toc}{chapter}{\listtablename}%
1796     \@mkboth{%
1797         \MakeUppercase\listtablename}%
1798         {\MakeUppercase\listtablename}%
1799     \@starttoc{lot}%
1800     \if@restonecol\twocolumn\fi
1801 }

\printlosymbols
1802 \newcommand\printlosymbols{%
1803 \renewcommand\glossaryname{\listsymbolname}%
1804 \@input@{\jobname.los}}

\makelosymbols
1805 \def\makelosymbols{%
1806     \newwrite\@losfile
1807     \immediate\openout\@losfile=\jobname.syx
1808     \newcommand\symb1[3][\@bsphack\begin{group}
1809     \ifstreampy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
1810         \@sanitize%
1811         \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
1812     {Writing index of symbols file \jobname.syx}%
1813     \let\makelosymbols\@empty%
1814 }%
1815 \@onlypreamble\makelosymbols

```

```

1816 \AtBeginDocument{%
1817 \ifpackageloaded{hyperref}{%
1818   \newcommand\@wrls[3]{%
1819     \protected@write\@losfile{%
1820       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1821     \endgroup%
1822     \@esphack}}{%
1823   \newcommand\@wrls[3]{%
1824     \protected@write\@losfile{%
1825       {\string\indexentry{#1[#2] #3}{\thepage}}%
1826     \endgroup%
1827     \@esphack}}}%

```

\printloabbreviations

```

1828 \newcommand\printloabbreviations{%
1829 \renewcommand\glossaryname{\listabbreviationname}%
1830 \@input@{\jobname.lab}}

```

\makeloabbreviations

```

1831 \def\makeloabbreviations{%
1832   \newwrite\@labfile
1833   \immediate\openout\@labfile=\jobname.abx
1834   \newcommand\abbrev[3][\@bsphack\beginingroup
1835     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
1836     \@sanitize
1837     \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
1838   {Writing index of abbreviations file \jobname.abx}%
1839   \let\makeloabbreviations\@empty
1840 }
1841 \@onlypreamble\makeloabbreviations

1842 \AtBeginDocument{%
1843 \ifpackageloaded{hyperref}{%
1844   \newcommand\@wrlab[3]{%
1845     \protected@write\@labfile{%
1846       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1847     \endgroup%
1848     \@esphack}}{%
1849   \newcommand\@wrlab[3]{%
1850     \protected@write\@labfile{%
1851       {\string\indexentry{#1[#2] #3}{\thepage}}%
1852     \endgroup%
1853     \@esphack}}}%

```

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
 \sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

1854 \newcommand\newsigla[3]{%
1855   \@namedef{coppe@sigla@s@#1}{#2}%
1856   \@namedef{coppe@sigla@l@#1}{#3}}
1857 \newcommand\coppe@sigla@reg[1]{%
1858   \@ifundefined{abbrev}{}}%

```

```

1859     {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}}
1860 \newcommand\coppe@sigla@full[1]{%
1861   \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
1862 \newcommand\coppe@sigla@undef[1]{%
1863   \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
1864   \textbf{???#1??}}
1865 \newcommand\coppe@sigla@first[1]{%
1866   \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
1867 \newcommand\coppe@sigla@norm[1]{%
1868   \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
1869   {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
1870    {\@nameuse{coppe@sigla@s@#1}}}}
1871 \newcommand\coppe@sigla@star[1]{%
1872   \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
1873   {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}}
1874 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

1875 %%% \AtBeginDocument{%
1876 %%% \ifpackageloaded{hyperref}{%
1877 %%%   \def\@wrlab#1#2{%
1878 %%%     \protected@write\@labfile{%
1879 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}}%
1880 %%%     \endgroup
1881 %%%     \@esphack}}{%
1882 %%%     \def\@wrlab#1#2{%
1883 %%%       \protected@write\@labfile{%
1884 %%%         {\string\indexentry{[#1] #2}{\arabic{page}}}}%
1885 %%%       \endgroup
1886 %%%       \@esphack}}}

1887 % Some macros used to generate cataloging information.
1888 \AtBeginDocument{%
1889   \ltx@ifpackageloaded{hyperref}{
1890     \def\coppe@bibEnd{%
1891       \immediate\write\@auxout{%
1892         \string\newlabel{bib:end}{\arabic{page}}{page.\arabic{page}}}}}%
1893     \def\coppe@bibBegin{%
1894       \immediate\write\@auxout{%
1895         \string\newlabel{bib:begin}{\arabic{page}}{page.\arabic{page}}}}}%
1896     \def\coppe@mainBegin{%
1897       \immediate\write\@auxout{%
1898         \string\newlabel{front:pageno}{\Roman{page}}{page.\roman{page}}}}}%
1899     \def\coppe@hasLof{%
1900       \immediate\write\@auxout{%
1901         \string\newlabel{cat:loflag}{~il.;}{page.\roman{page}}}}}%
1902   }{%
1903     \def\coppe@bibEnd{%
1904       \immediate\write\@auxout{%
1905         \string\newlabel{bib:end}{\arabic{page}}}}}%
1906     \def\coppe@bibBegin{%
1907       \immediate\write\@auxout{%
1908         \string\newlabel{bib:begin}{\arabic{page}}}}}%
1909     \def\coppe@mainBegin{%
1910       \immediate\write\@auxout{%
1911         \string\newlabel{front:pageno}{\Roman{page}}}}}%

```

```

1912 \def\coppe@hasLof{%
1913 \immediate\write\@auxout{%
1914 \string\newlabel{cat:lofflag}{\{:\sim il.;\}}}%
1915 }%
1916 }
1917 % The reference list is now produced by biblatex's \printbibliography (see the
1918 % bibliography setup after babel, above). The former thebibliography
1919 % redefinition and its \bibindent are no longer used and have been removed; the
1920 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
1921 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

1922 \newlength{\recuolongquote}%
1923 \setlength{\recuolongquote}{4cm}%
1924 \newenvironment*{longquote}[1][default]{%
1925 \list{}%
1926 \footnotesize%
1927 \addtolength{\leftskip}{\recuolongquote}%
1928 \item[]%
1929 \singlespacing%
1930 \ifthenelse{not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
1931 }\endlist}%

1932 \newenvironment{theglossary}{%
1933 \if@twocolumn%
1934 \@restonecoltrue\onecolumn%
1935 \else%
1936 \@restonecolfalse%
1937 \fi%
1938 \@mkboth{\MakeUppercase\glossaryname}%
1939 {\MakeUppercase\glossaryname}%
1940 \chapter*{\glossaryname}%
1941 \addcontentsline{toc}{chapter}{\glossaryname}
1942 \list{}
1943 {\setlength{\listparindent}{0in}%
1944 \setlength{\labelwidth}{1.0in}%
1945 \setlength{\leftmargin}{1.5in}%
1946 \setlength{\labelsep}{0.5in}%
1947 \setlength{\itemindent}{0in}}%
1948 \sloppy}%
1949 {\if@restonecol\twocolumn\fi%
1950 \endlist}
1951 %
1952 \renewenvironment{theindex}{%
1953 \if@twocolumn
1954 \@restonecolfalse
1955 \else
1956 \@restonecoltrue
1957 \fi
1958 \twocolumn[\@makeschapterhead{\indexname}]%
1959 \@mkboth{\MakeUppercase\indexname}%
1960 {\MakeUppercase\indexname}%
1961 \thispagestyle{plain}\parindent\z@
1962 \addcontentsline{toc}{chapter}{\indexname}

```

```

1963 \parskip\z@ \@plus .3\p@\relax
1964 \columnseprule \z@
1965 \columnsep 35\p@
1966 \let\item\@idxitem}
1967 {\if@restonecol\onecolumn\else\clearpage\fi}
1968 \if@english
1969 \newcommand\listabbreviationname{List of Abbreviations and Acronyms}
1970 \newcommand\listsymbolname{List of Symbols}
1971 \newcommand\glossaryname{Glossary}
1972 \else
1973 \newcommand\listabbreviationname{Lista de Abreviaturas e Siglas}
1974 \newcommand\listsymbolname{Lista de S{\` i}mbolos}
1975 \newcommand\glossaryname{Gloss{\` a}rio}
1976 \fi
1977 %
1978 \newcommand\local@advisorstring{Orientador}
1979 \newcommand\foreign@advisorstring{Advisor}
1980 \ifthenelse{\boolean{maledoc}}{%
1981 \newcommand\local@approvedname{Examinado por}%
1982 }{%
1983 \newcommand\local@approvedname{Examinada por}%
1984 }
1985 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
1986 \newcommand\local@deptstring{Programa}
1987 \newcommand\foreign@deptstring{Department}
1988 \newcommand\local@cityname{Rio de Janeiro}
1989 \newcommand\local@statename{RJ}
1990 \newcommand\local@countryname{Brasil}
1991 %
1992 \newcommand\frontcover@maintext{
1993 \sloppy\nohyphens{\local@doctype\ de \@degreename\
1994 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1995 ao Programa de P{\` o}S-gradua{\c c}{\~ a}o em \local@deptname,
1996 COPPE, da \local@universityname, como parte dos requisitos
1997 necess{\` a}rios {\` a} obten{\c c}{\~ a}o do t{\` i}tulo de
1998 \local@degname\ em \local@deptname.}
1999 }
2000 %
2001 \newcommand\frontpage@maintext{
2002 \noindent {\MakeUppercase\local@doctype}
2003 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2004 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2005 DE P{\` O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2006 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2007 NECESS{\` A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
2008 {\MakeUppercase\local@degname} EM CI{\` E}NCIAS EM
2009 {\MakeUppercase\local@deptname.\par}}%
2010 }
2011 %
2012 \newcommand\frontpage@bottomtext{%
2013 \begin{center}
2014 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2015 {\MakeUppercase\local@monthname\ DE \number\year}
2016 \end{center}}%

```

```

2017 }
2018 %
2019 \newcommand\abstract@toptext{%
2020 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
2021 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2022 \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
2023 necess{\` a}rios para a obten{\c c}{\~ a}o do grau de
2024 \local@degname\ em Ci{\^ e}ncias (\@degree)}
2025 }
2026 \newcommand\foreignabstract@toptext{%
2027 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2028 COPPE/UFRJ as a partial fulfillment of the requirements for the
2029 degree of \foreign@degname\ of Science (\@degree)}
2030 }
2031 %
2032 </class>
2033 <*glossary>
2034 actual '='
2035 quote '!'
2036 level '>'
2037 %%% delim_0 ", p. "
2038 delim_0 "\\dotfill "
2039 lethead_flag 0
2040 headings_flag 0
2041 preamble
2042 "\n\\begin{theglossary}\n \\makeatletter"
2043 postamble
2044 "\n \\end{theglossary}\n"
2045 </glossary>
2046 <*class>
2047 \newcommand{\annex}{
2048 \iflanguage{brazilian}
2049 {\renewcommand{\appendixname}{Anexo}}
2050 {\renewcommand{\appendixname}{Annex}}
2051 \appendix
2052 \@coppeannextrue
2053 }
2054 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

2055 <*dbx>
2056 \ProvidesFile{coppe.dbx}[2026/05/23 v3.8 CoppeTeX ABNT datamodel]
2057 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
2058 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
2059 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
2060 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
2061 % performance) are already in biblatex's standard datamodel.
2062 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
2063 % Custom fields.
2064 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"

```

```

2065 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
2066 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
2067 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
2068 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
2069 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
2070 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
2071 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
2072 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
2073 \DeclareDatamodelEntryfields{course,coppedegree}
2074 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
2075 {artist,platform,director,producer,version}
2076 \DeclareDatamodelEntryfields[map]{scale}
2077 \DeclareDatamodelEntryfields[article]{caderno}
2078 \DeclareDatamodelEntryfields[jurisdiction]{relator}
2079 </dbx>
2080 <*bbx>
2081 \ProvidesFile{coppe.bbx}[2026/05/23 v3.8 CoppeTeX ABNT bibliography style]
2082 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
2083 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
2084 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
2085 % so load setspace here as well (idempotent).
2086 \RequirePackage{setspace}
2087 % ABNT reference list keeps the date at the end (imprenta), same for author-date
2088 % and numeric modes, so we derive from the standard style, not authoryear.
2089 \RequireBibliographyStyle{standard}
2090
2091 % Note: standard style never dashes repeated authors, which already matches the
2092 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
2093 \ExecuteBibliographyOptions{%
2094   maxbibnames=99, % ABNT 4.3.1.3: list all authors by default
2095   giveninits=false,
2096   uniquename=false,
2097   uniquelist=false,
2098   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
2099   sorting=nyt,
2100 }
2101
2102 % Language strings (Disponível em / Acesso em / In, etc.)
2103 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
2104 \DeclareLanguageMapping{english}{english-coppe}
2105
2106 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
2107 \DeclareNameAlias{author}{family-given}
2108 \DeclareNameAlias{editor}{family-given}
2109 \DeclareNameAlias{translator}{family-given}
2110 \DeclareNameAlias{default}{family-given}
2111 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
2112 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
2113
2114 % Uppercase the surname (family + prefix) only inside the bibliography list;
2115 % citations keep normal caps (post-2020 rule).
2116 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
2117 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
2118 % field names; the month stays abbreviated via \mkbibmonth.

```

```

2119 \protected\def\coppe@abntdate#1#2#3{%
2120   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
2121   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
2122   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}%
2123 \AtBeginBibliography{%
2124   \singlespacing % entries single-spaced internally (2.4)
2125   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
2126   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
2127   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
2128   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
2129   \let\mkbibdateshort\coppe@abntdate
2130 }
2131
2132 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
2133 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
2134 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
2135 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
2136 % article AND the next word are uppercased; otherwise just the first word. The
2137 % delimited macros split off words; \coppeUCkeep strips the trailing-space
2138 % sentinel so no spurious space is left before the following punctuation.
2139 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
2140 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
2141   \coppeUCifart{#1}
2142   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
2143   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
2144 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
2145   \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
2146 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
2147 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
2148 \protected\def\coppeUCifart#1{%
2149   \begingroup
2150   \lowercase{\def\coppeUC@w{#1}}%
2151   \edef\coppeUC@w{\coppeUC@w}%
2152   \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
2153   \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
2154   \endgroup}
2155 \DeclareFieldFormat{title}{%
2156   \ifboolexpr{ test {\ifnameundef{labelname}} }
2157     {#1}%
2158     {\mkbibbold{#1}}}
2159 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
2160 % format receives the RAW title text (the title format above only sees the
2161 % field-printing commands), so \coppeUCtitle can split off and uppercase the
2162 % first word here.
2163 \DeclareFieldFormat{titlecase}{%
2164   \ifboolexpr{ test {\ifnameundef{labelname}} }
2165     {\coppeUCtitle{#1}}%
2166     {#1}}
2167 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
2168 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
2169 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
2170 % the title of a whole document, so override those back to bold.
2171 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
2172 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT

```



```

2173 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
2174 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
2175 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
2176 \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
2177 \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
2178
2179 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
2180 \DeclareFieldFormat[pagetotal]{#1\addnspace p\adddot}
2181 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnspace f\adddot}
2182 \DeclareFieldFormat[pages]{p\adddot\addnspace#1}
2183
2184 % --- Periodical volume / number prefixes ---
2185 \DeclareFieldFormat[article]{volume}{v\adddot\addnspace#1}
2186 \DeclareFieldFormat[article]{number}{n\adddot\addnspace#1}
2187
2188 % --- "In:" before host title (incollection/inproceedings/inbook) ---
2189 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
2190
2191 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
2192 \DeclareFieldFormat[url]{\bibstring{availablefrom}\addcolon\space\url{#1}}
2193
2194 % --- ABNT article driver: no "In."; Journal, Local, v., n., p., date, DOI/URL.
2195 %     Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
2196 %     differ: the newspaper name (journaltitle) and city, then either
2197 %     "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
2198 %     "p. <pages>, <date>" (page before date, when none),
2199 %     per NBR 6023 §4.2.3.6. ---
2200 \DeclareBibliographyDriver{article}{%
2201   \usebibmacro{bibindex}%
2202   \usebibmacro{begentry}%
2203   \usebibmacro{author/translator+others}%
2204   \setunit{\labelnamepunct}\newblock
2205   \usebibmacro{title}%
2206   \newunit\newblock
2207   \printfield{journaltitle}%
2208   \setunit{\addcomma\space}\printlist{location}%
2209   \iffieldequalstr{entrysubtype}{newspaper}
2210     {\iffieldundef{caderno}
2211       {\setunit{\addcomma\space}\printfield{pages}%
2212         \setunit{\addcomma\space}\printdate}%
2213       {\setunit{\addcomma\space}\printdate%
2214         \setunit{\adddot\space}\printfield{caderno}%
2215         \setunit{\addcomma\space}\printfield{pages}}}%
2216   {\setunit{\addcomma\space}\printfield{volume}%
2217     \setunit{\addcomma\space}\printfield{number}%
2218     \setunit{\addcomma\space}\printfield{pages}%
2219     \setunit{\addcomma\space}\printdate}%
2220   \newunit\newblock
2221   \printfield{doi}%
2222   \newunit\newblock
2223   \usebibmacro{url+urldate}%
2224   \usebibmacro{finentry}}
2225
2226 % --- incollection: ABNT puts the host organizer before the booktitle, as

```

```

2227 %      "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
2228 %      p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
2229 \newbibmacro*{coppe:bookeditor}{%
2230   \ifnameundef{editor}
2231   {}
2232   {\printnames{editor}%
2233    \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2234    \clearname{editor}%
2235    \setunit{\adddot\space}}
2236 \newbibmacro*{coppe:partdriver}{%
2237   \usebibmacro{bibindex}\usebibmacro{begentry}%
2238   \usebibmacro{author/translator+others}%
2239   \setunit{\labelnamepunct}\newblock
2240   \usebibmacro{title}%
2241   \newunit\newblock
2242   \usebibmacro{in:}%
2243   \usebibmacro{coppe:bookeditor}%
2244   \printfield{booktitle}%
2245   \newunit\newblock
2246   \printfield{edition}%
2247   \newunit\newblock
2248   \printlist{location}%
2249   \setunit*{\addcolon\space}\printlist{publisher}%
2250   \setunit*{\addcomma\space}\printdate%
2251   \newunit\newblock
2252   \printfield{pages}%
2253   \setunit{\addcomma\space}\printfield{chapter}%
2254   \newunit\newblock
2255   \usebibmacro{url+urldate}%
2256   \usebibmacro{finentry}}
2257 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
2258 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
2259
2260 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
2261 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
2262
2263 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
2264 \DeclareFieldFormat{course}{\mkbibparens{\iflanguage{brazilian}{em\addspace}{in\addspace}#1}}
2265 \DeclareBibliographyDriver{thesis}{%
2266   \usebibmacro{bibindex}\usebibmacro{begentry}%
2267   \usebibmacro{author}%
2268   \setunit{\labelnamepunct}\newblock
2269   \usebibmacro{title}%
2270   \newunit\newblock
2271   \printfield{type}\setunit*{\addspace}\printfield{course}%
2272   \newunit\newblock
2273   \printlist{institution}%
2274   \setunit*{\addcomma\space}\printlist{location}%
2275   \setunit*{\addcomma\space}\printdate%
2276   \newunit\newblock
2277   \printfield{pagetotal}%
2278   \newunit\newblock
2279   \printfield{note}%
2280   \newunit\newblock

```

```

2281 \usebibmacro{url+urldate}%
2282 \usebibmacro{finentry}}
2283 % --- Games / audiovisual: render the custom fields (artist, platform,
2284 % director, producer, version) the generic misc driver ignores. ---
2285 \newbibmacro*{coppe:media}{%
2286 \iflistundef{artist}{\setunit{\addperiod\space}%
2287 \printtext{\iflanguage{brazilian}{Ilustra\c{c}\~ao}{Art}\addcolon\space}\printlist{artist}}
2288 \iffieldundef{platform}{\setunit{\addperiod\space}%
2289 \printtext{\iflanguage{brazilian}{Plataforma}{Platform}\addcolon\space}\printfield{platform}}
2290 \iflistundef{director}{\setunit{\addperiod\space}%
2291 \printtext{\iflanguage{brazilian}{Dire\c{c}\~ao}{Directed by}\addcolon\space}\printlist{director}}
2292 \iflistundef{producer}{\setunit{\addperiod\space}%
2293 \printtext{\iflanguage{brazilian}{Produ\c{c}\~ao}{Produced by}\addcolon\space}\printlist{producer}}
2294 \newbibmacro*{coppe:mediadriv}{%
2295 \usebibmacro{bibindex}\usebibmacro{begentry}%
2296 \usebibmacro{author/translator+others}%
2297 \setunit{\labelnamepunct}\newblock
2298 \usebibmacro{title}%
2299 \newunit\newblock
2300 \usebibmacro{coppe:media}%
2301 \newunit\newblock
2302 \printlist{publisher}%
2303 \setunit*{\addcomma\space}\printlist{location}%
2304 \setunit*{\addcomma\space}\printdate%
2305 \newunit\newblock
2306 \printfield{version}%
2307 \newunit\newblock
2308 \printfield{note}%
2309 \newunit\newblock
2310 \usebibmacro{url+urldate}%
2311 \usebibmacro{finentry}}
2312 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriv}}
2313 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriv}}
2314 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriv}}
2315 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriv}}
2316 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriv}}
2317 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2318 % imprenta) plus the mandatory scale ("Escala 1:N").
2319 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
2320 \DeclareBibliographyDriver{map}{%
2321 \usebibmacro{bibindex}\usebibmacro{begentry}%
2322 \usebibmacro{author/translator+others}%
2323 \setunit{\labelnamepunct}\newblock
2324 \usebibmacro{title}%
2325 \newunit\newblock
2326 \printlist{location}%
2327 \setunit*{\addcolon\space}\printlist{publisher}%
2328 \setunit*{\addcomma\space}\printdate%
2329 \newunit\newblock
2330 \printfield{scale}%
2331 \newunit\newblock
2332 \usebibmacro{url+urldate}%
2333 \usebibmacro{finentry}}
2334 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type

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2335 % number. date. The holder (depositante) is printed in normal case, not the
2336 % ALL-CAPS family treatment the bibliography gives the leading author.
2337 \newbibmacro*{coppe:depositor}{%
2338   \ifnameundef{holder}
2339     {}
2340     {\printtext{\bibstring{depositor}\addcolon\space}%
2341       \begingroup
2342         \def\mkbibnamefamily##1{##1}%
2343         \def\mkbibnameprefix##1{##1}%
2344         \printnames{holder}%
2345       \endgroup}}
2346 \DeclareBibliographyDriver{patent}{%
2347   \usebibmacro{bibindex}%
2348   \usebibmacro{begentry}%
2349   \usebibmacro{author/translator+others}%
2350   \setunit{\labelnamepunct}\newblock
2351   \usebibmacro{title}%
2352   \newunit\newblock
2353   \usebibmacro{coppe:depositor}%
2354   \newunit\newblock
2355   \printfield{type}%
2356   \setunit*{\addspace}\printfield{number}%
2357   \newunit\newblock
2358   \printdate%
2359   \newunit\newblock
2360   \usebibmacro{url+urldate}%
2361   \usebibmacro{finentry}}
2362 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2363 % give each a baseline (book-like for published docs: norma/partitura; misc for
2364 % the rest) so they print. Per-type ABNT refinements can follow later.
2365 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2366 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
2367 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2368 % the law id + ementa (plain, not bold), then location and date.
2369 \newbibmacro*{coppe:legaldriver}{%
2370   \usebibmacro{bibindex}\usebibmacro{begentry}%
2371   \usebibmacro{author/translator+others}%
2372   \setunit{\labelnamepunct}\newblock
2373   \usebibmacro{title}%
2374   \newunit\newblock
2375   % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
2376   \iffielddundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
2377   \printlist{location}%
2378   \setunit{\addcomma\space}\printdate%
2379   \newunit\newblock
2380   \usebibmacro{url+urldate}%
2381   \usebibmacro{finentry}}
2382 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2383 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
2384 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
2385 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2386 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2387 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2388 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}

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2389 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2390 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2391 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2392
2393 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2394 \DeclareSourcemap{%
2395   \maps[datatype=bibtex]{%
2396     % entry-type PT synonyms -> canonical English
2397     \map{\step[tourcesource=livro,typetarget=book]}
2398     \map{\step[tourcesource=artigo,typetarget=article]}
2399     \map{\step[tourcesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2400     \map{\step[tourcesource=partedelivro,typetarget=incollection]}
2401     \map{\step[tourcesource=emlivro,typetarget=inbook]}
2402     \map{\step[tourcesource=verbete,typetarget=inreference]}
2403     \map{\step[tourcesource=anais,typetarget=proceedings]}
2404     \map{\step[tourcesource=trabalhodeevento,typetarget=inproceedings]}
2405     \map{\step[tourcesource=periodico,typetarget=periodical]}
2406     \map{\step[tourcesource=relatorio,typetarget=report]}
2407     \map{\step[tourcesource=relatoriotecnico,typetarget=report]}
2408     \map{\step[tourcesource>manual,typetarget>manual]}
2409     \map{\step[tourcesource=correspondencia,typetarget=letter]}
2410     \map{\step[tourcesource=patente,typetarget=patent]}
2411     \map{\step[tourcesource=diverso,typetarget=misc]}
2412     \map{\step[tourcesource=jogo,typetarget=game]}
2413     \map{\step[tourcesource=jogoeletronico,typetarget=videogame]}
2414     \map{\step[tourcesource=jogodetabuleiro,typetarget=boardgame]}
2415     \map{\step[tourcesource=filme,typetarget=movie]}
2416     \map{\step[tourcesource=serietv,typetarget=tvshow]}
2417     \map{\step[tourcesource=programadetelevisao,typetarget=tvshow]}
2418     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2419     % except map (declared in coppe.dbx).
2420     \map{\step[tourcesource=legislacao,typetarget=legislation]}
2421     \map{\step[tourcesource=atnormativo,typetarget=legislation]}
2422     \map{\step[tourcesource=jurisprudencia,typetarget=jurisdiction]}
2423     \map{\step[tourcesource=documentojuridico,typetarget=legal]}
2424     \map{\step[tourcesource=documentocivil,typetarget=legal]}
2425     \map{\step[tourcesource=norma,typetarget=standard]}
2426     \map{\step[tourcesource=partitura,typetarget=music]}
2427     \map{\step[tourcesource=iconografico,typetarget=image]}
2428     \map{\step[tourcesource=obradearte,typetarget=artwork]}
2429     \map{\step[tourcesource=tridimensional,typetarget=artwork]}
2430     \map{\step[tourcesource=cartografico,typetarget=map]}
2431     \map{\step[tourcesource=mapa,typetarget=map]}
2432     \map{\step[tourcesource=audiovisual,typetarget=video]}
2433     \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
2434     \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}
2435     \map{\step[tourcesource=trabalhoapresentado,typetarget=unpublished]}
2436     % thesis family: retype to thesis and preset the (babel-aware) type word
2437     \map{\step[tourcesource=tese,typetarget=thesis]}
2438     \map{\step[tourcesource=dissertacao,typetarget=thesis]}
2439     \map{\step[tourcesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2440     \map{\step[tourcesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2441     \map{\step[tourcesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2442     \map{\step[tourcesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d

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2443 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
2444 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2445 \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2446 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
2447 % field PT synonyms (unaccented) -> canonical English
2448 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
2449 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
2450 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
2451 \map{\step[fieldsource=titulo,fieldtarget=title]}
2452 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
2453 \map{\step[fieldsource=autor,fieldtarget=author]}
2454 \map{\step[fieldsource=editora,fieldtarget=publisher]}
2455 \map{\step[fieldsource=local,fieldtarget=location]}
2456 \map{\step[fieldsource=ano,fieldtarget=year]}
2457 \map{\step[fieldsource=data,fieldtarget=date]}
2458 \map{\step[fieldsource=edicao,fieldtarget=edition]}
2459 \map{\step[fieldsource=paginas,fieldtarget=pages]}
2460 \map{\step[fieldsource=numero,fieldtarget=number]}
2461 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
2462 \map{\step[fieldsource=organizador,fieldtarget=editor]}
2463 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
2464 \map{\step[fieldsource=tipo,fieldtarget=type]}
2465 \map{\step[fieldsource=nota,fieldtarget=note]}
2466 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
2467 \map{\step[fieldsource=curso,fieldtarget=course]}
2468 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
2469 \map{\step[fieldsource=artista,fieldtarget=artist]}
2470 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
2471 \map{\step[fieldsource=diretor,fieldtarget=director]}
2472 \map{\step[fieldsource=produtor,fieldtarget=producer]}
2473 \map{\step[fieldsource=versao,fieldtarget=version]}
2474 \map{\step[fieldsource=escala,fieldtarget=scale]}
2475 }%
2476 }
2477
2478 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2479 \setlength{\bibhang}{0pt}
2480 \setlength{\bibitemsep}{\baselineskip}
2481
2482 </bbx>
2483 <*cbx>
2484 \ProvidesFile{coppe.cbx}[2026/05/23 v3.8 CoppeTeX ABNT author-date citations]
2485 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2486 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2487 \RequireCitationStyle{authoryear-comp}
2488
2489 % --- NBR 10520:2023 author separator -----
2490 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2491 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
2492 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2493 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2494 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2495 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2496 \DeclareDelimFormat{textcite}{multinamedelim}{\addcomma\space}

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2497 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2498
2499 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2500 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
2501 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2502 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2503 % so it is never added to the bibliography.
2504 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2505 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2506 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2507 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2508 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2509 \DeclareRobustCommand{\citetapud}[4][]{%
2510 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
2511 \DeclareRobustCommand{\citepapud}[4][]{%
2512 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2513 \citeauthor{#4}\addcomma\space\citeyear{#4}%
2514 \ifblank{#1}{ }\{\addcomma\space\mkpageprefix[pagination]{#1}\}}
2515 \</cbx>
2516 \<*\bxbr>
2517 \ProvidesFile{brazilian-coppe.lbx}[2026/05/23 v3.8 CoppeTeX brazilian strings]
2518 \InheritBibliographyExtras{brazilian}
2519 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2520 \DeclareBibliographyStrings{%
2521 inherit = {brazilian},
2522 availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
2523 urlseen = {{Acesso em}{Acesso em}},
2524 in = {{In}{In}},
2525 editor = {{Organizador}{Org\addot}},
2526 editors = {{Organizadores}{Orgs\addot}},
2527 depositor = {{Depositante}{Depositante}},
2528 coppescale = {{Escala}{Escala}},
2529 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2530 dscthesi = {{Tese de Doutorado}{Tese de Doutorado}},
2531 phdthesis = {{Tese de Doutorado}{Tese de Doutorado}},
2532 tcc = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2533 }
2534 \</\bxbr>
2535 \<*\bxen>
2536 \ProvidesFile{english-coppe.lbx}[2026/05/23 v3.8 CoppeTeX english strings]
2537 \InheritBibliographyExtras{english}
2538 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2539 \DeclareBibliographyStrings{%
2540 inherit = {english},
2541 availablefrom = {{Available from}{Available from}},
2542 urlseen = {{Access on}{Access on}},
2543 in = {{In}{In}},
2544 depositor = {{Depositor}{Depositor}},
2545 coppescale = {{Scale}{Scale}},
2546 mscdiss = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
2547 dscthesi = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
2548 phdthesis = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
2549 tcc = {{Undergraduate Thesis}{Undergraduate Thesis}},
2550 }

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```

2551 </lboxen>
2552 <*numbbx>
2553 \ProvidesFile{coppe-numeric.bbx}[2026/05/24 v3.8 CoppeTeX ABNT numeric bibliography]
2554 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
2555 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
2556 % the PT->EN source map and the language mappings) and only adds the [n] list
2557 % label and the numeric sorting on top.
2558 \RequireBibliographyStyle{coppe}
2559
2560 % ABNT numeric system lists references in the order they are first cited
2561 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
2562 % numbered list. 'labelnumber' generates the [n] field.
2563 \ExecuteBibliographyOptions{labelnumber,sorting=none}
2564
2565 % Bracketed labels, both in the list and for any shorthand entries.
2566 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
2567 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
2568
2569 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
2570 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
2571 \defbibenvironment{bibliography}
2572 {
\list
2573   {\printtext[labelnumberwidth]{%
2574     \printfield{labelprefix}%
2575     \printfield{labelnumber}}}
2576   {\setlength{\labelwidth}{\labelnumberwidth}%
2577     \setlength{\leftmargin}{\labelwidth}%
2578     \setlength{\labelsep}{\biblabelsep}%
2579     \addtolength{\leftmargin}{\labelsep}%
2580     \setlength{\itemsep}{\bibitemsep}%
2581     \setlength{\parsep}{\bibparsep}}%
2582   \renewcommand*{\makelabel}[1]{\hss##1}}
2583 {\endlist}
2584 {\item}
2585
2586 </numbbx>
2587 <*numcbx>
2588 \ProvidesFile{coppe-numeric.cbx}[2026/05/24 v3.8 CoppeTeX ABNT numeric citations]
2589 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
2590 % numbers [1], compressed and sorted within a single cite group. This is the
2591 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
2592 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
2593 % \citet -> Author [1].
2594 \RequireCitationStyle{numeric-comp}
2595 </numcbx>

```

Acknowledgments

Thanks to all COPPET_EX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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\,	 783, 792	 1075, 1076
\;	 723, 725, 727	\@ifstar 1874
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